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Removal of Paint on Al Alloy Substrate without Destruction by Air Jet with Ultrahard Ice Particles

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Abstract
Aluminum (Al) alloy is widely used and coated with a protective layer. When the layer ages, it needs to be removed before re-coating, but conventional methods frequently cause substrate damage, environmental contamination, or require secondary cleaning. While ice particle jets offer a promising alternative, existing approaches using micron-sized or low-hardness ice particles often lack sufficient kinetic energy for efficient cleaning. In this paper, the air jet technology with ultrahard ice particles as abrasives (AJUIP) was utilized to experimentally investigate the removal of epoxy organic silicon paint layer from the surface of Al alloy. The results showed that, driven by high-speed air, the ice particles could effectively remove the paint layer from the substrate without causing any damage to the substrate. By optimizing cleaning processing parameters, the efficiency and quality of the cleaning could be improved. The study established a four-stage removal process (nucleation, growth, fusion, cleaning) characterized by circular paint peeling units. Elemental analysis and scanning electron microscopy images revealed nearly no residual paint in the cleaned area. Numerical simulations of the air flow field elucidated the asymmetric gas velocity distribution influencing the distinct morphologies of the removal region boundaries. Finally, an elastic contact theory explanation for the paint removal mechanism of the AJUIP was provided, i.e., the size and velocity of the ice particles were the main factors affecting the paint removal effect. Therefore, the AJUIP exhibited significant potential for application in surface cleaning.
Keywords: Paint removal, Cleaning technology, Ice jet, Aluminum alloy

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1 Introduction

Aluminum (Al) alloy is a lightweight material with excellent properties [1]. Replacing traditional metals with it can reduce component weight while maintaining mechanical performance. It's widely used in fields like ship [2], automotive [3], aerospace [4, 5], and architecture. To improve its surface performance in extreme environments, various surface treatment technologies have been developed, such as anodic oxidation [6], passivation treatment [7], and painting. Painting treatment is common for rust prevention [8], waterproofing [9, 10], corrosion resistance [11, 12], and aesthetics, but the paint layer on the Al alloy surface deteriorates over time, causing cracking and peeling, thus requiring regular repainting [13, 14]. Before a new painting, it is essential to remove the residual aged paint layer in order to ensure the consistency of the fresh paint layer, and the quality of paint removal has a direct impact on the subsequent use and performance of equipment.

Current paint removal methods include abrasive jet [15, 16], mechanical polishing, laser paint removal [17], plasma paint removal [18], and chemical paint removers [19]. However, these methods all have their own limitations. Mechanical polishing exhibits a low rate of paint removal, and the operators are exposed to excessive work intensity and susceptible to dust invasion. Additionally, it is also easy to cause a damage to the substrate. Chemical paint removers may corrode the metal surface, and the resulting waste liquid may release toxic gases, posing a threat to the health of maintenance personnel and the environment. Laser and plasma paint removals represent advanced and effective methods for eliminating paint, yet these techniques may result in the deposition of ash or metal oxides on the substrate surface following the combustion of the residual paint layer. Consequently, supplementary cleaning procedures are necessary to ensure their complete eradication. Abrasive jet technique represents an efficient and energy-efficient method for paint removal. It utilizes high-speed flowing jet media, such as high-pressure air or high-pressure water, to accelerate abrasives (e.g., garnet, alumina, quartz sand, and so on) and achieve paint removal by impacting the workpiece with abrasive particles. Nevertheless, conventional abrasives

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pose challenges in terms of recovery, and excessively sharp abrasives often cause damage to the substrate surface during paint removal. For example, Yang et al. carried out a sandblasting paint removal process on the paint layer of aircraft skins using plastic particles [20]. The results indicated that the abrasive particles would leave a considerable number of impact craters on the substrate surface and even lead to the warping of the specimens, which seriously affected the subsequent use of the components.

In contrast to the aforementioned abrasive jets, the ice particle jet cleaning technology emerges as a novel solution for paint removal, substituting conventional mineral abrasives with ice particles. When colliding with the workpiece, the ice particles transform into water, effectively evading the environmental contamination tied to traditional abrasives [21, 22]. In addition, water, the product of ice, can also carry away the peeled-off paint layer. Even more significantly, distinct from the traditional abrasives, the hardness of ice particles can be modulated by varying their temperature [23], allowing it to lie between the hardness of the substrate and the paint layer, thereby attaining the cleaning of the substrate without causing any damage. Li et al. [21] used −30 °C ice particles to remove the acrylic paint layer on 2A12 aluminum alloy surface. They found that the ice air jet had a good de-painting effect, and the surface roughness of the substrate changed little after blasting. Prezhbilov et al. [22] used a water-ice mixed jet to clean Polyurethane coatings and Automotive enamel and also achieved excellent effects.

Although there are existing reports about paint removal using the ice particle jet, there remain several limitations that need to be addressed. For instance, the water jet with ice particles can result in an increase in temperature of the abrasives, leading to a reduction in hardness and limitation of mechanical performance. Additionally, this method results in a significant waste of water resources during processing. Although the air jet with ice particles does not suffer from the aforementioned drawbacks, the ice particles utilized in current technology are tiny (micron-sized) and low-hardness, making it challenging to generate great kinetic energy for cleaning. In addition, the majority of paint removal experiments are carried out under laboratory conditions, and the paint layer samples used do not meet the necessary level of adhesion strength with the substrate for practical applications.

In response to the aforementioned concerns, this study proposed to conduct paint removal experiments on the heat-resistant epoxy organosilicone paint sprayed on the surface of 6061 Al alloy by utilizing the ultrahard ice particles with the Mohs hardness of 3.5 at −86 °C, without causing damage to the substrate. The study aimed to explore the influence of various processing parameters on the paint removal effect. Furthermore, in order to investigate the removal mechanism of the air jet with ultrahard ice particles on the paint layer, a finite element simulation method was employed to simulate air velocity near the nozzle and explore how the size and speed of the ice particle affect the paint layer damage. This research provided practical reference for improving the processing efficiency of the air jet with ice particles, precise removal of paint materials, and cleaning of substrates without destruction.

2 Experimental Details

2.1 Paint Removal System Based on the Air Jet with Ultrahard Ice Particles

The paint removal system based on the air jet with ultrahard ice particles (AJUIP) was composed of a high-pressure air supply unit, an ice blasting unit, and a motion control unit, as shown in Figure 1.

The high-pressure gas supply unit comprises an air compressor, a gas container, a refrigerated dryer, and three filters. The air compressor was capable of delivering compressed air at a maximum pressure of 0.8 MPa. However, in accordance with the principle of thermodynamics, the temperature of the compressed air would rise, potentially heating up the ice particles and reducing their hardness. Therefore, it was necessary to utilize a refrigerated dryer to cool down the high-pressure air and filter out its moisture. The filtered high-pressure air was then mixed with the ice particles in the ice jet cleaner to form the AJUIP, which was subsequently accelerated and sprayed out through the blast nozzle to impact the surface of the workpiece. The position and velocity of the nozzle were regulated by the motor-controlled motion platform in the motion control unit.

2.2 Preparation of Samples

6061 aluminum (Al) alloy is a high-performance material suitable for aircraft manufacturing, and the samples with dimensions of 50 mm×100 mm×1.5 mm were employed as substrates. To improve the hardness, wear resistance, heat resistance, and electrical insulation of the substrate, sulfuric acid was utilized to conduct anodizing treatment on the surface of the Al alloy, and then a hard anodic oxide layer with approximately 30 μm thickness was formed. The anodizing process

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adhered to the requirements outlined in the Chinese standard of "HB/Z 233-1993".

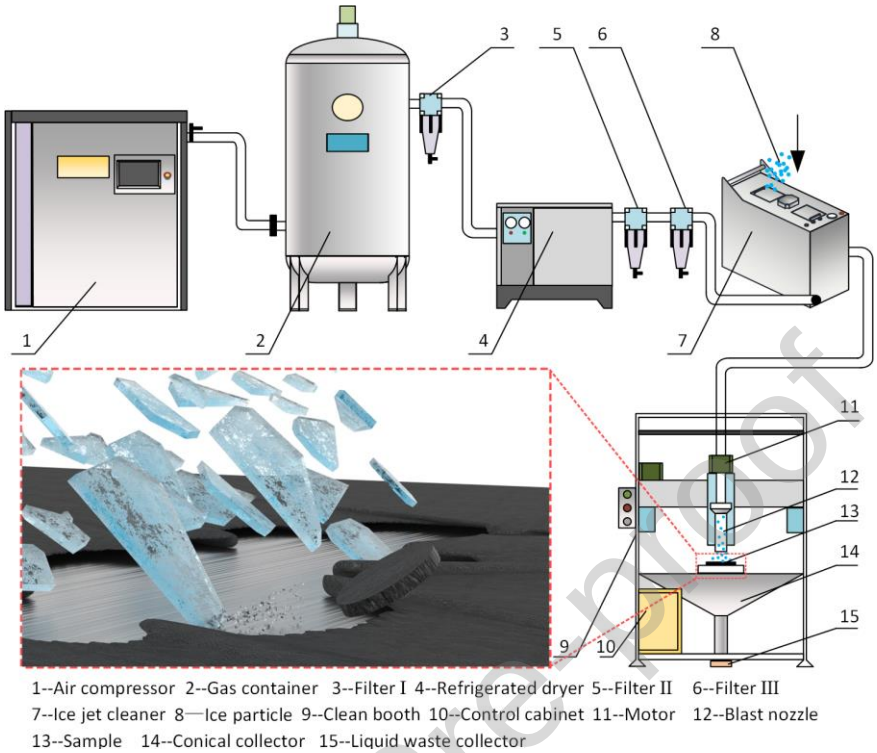

Figure 1 Schematic diagram of the paint removal system based on the AJUIP.

The paint of the samples featured a double-layer structure: a heat-resistant epoxy organosilicone was utilized as the primer coating to enhance the flatness of the metal surface and improve adhesion with the anodic oxide layer, while a heat-resistant epoxy silicone varnish served as the topcoat, providing fire resistance, moisture resistance, and pollution resistance. The spraying process and performance of these coatings followed the Chinese standards of "GJB 1591-1993" and "GJB 1592-1993".

2.3 Characterization

The microstructure of the paint layer and substrate was examined using a scanning electron microscope (SEM, Supra55, Zeiss Germany). The surface roughness and height profiles of the samples were quantified using a 3D surface optical profiler (NewView9000, ZYGO Corporation, America). The chemical elements of the sample surface were analyzed by energy dispersive X-ray spectroscopy (EDS, Ultim Max, Oxford Instruments, UK).

3 Results and Discussions

3.1 Characterization of Samples and Ice Particles

Figure 2(a) shows the macroscopic image of the prepared sample, which appeared to be a flat black surface devoid of any cracks or wrinkles. In Figure 2(b), the microscopic views of the paint layer surface revealed its rough texture, with many varying-sized paint agglomerates dispersed on the surface. Additionally, Figure 2(c) present the microscopic images of the cross-section of the sample after being encapsulated with epoxy resin, obviously illustrating its layered structure. The thicknesses of the topcoat and the primer were 25-30 μm and 35-40 μm, respectively. Furthermore, it was evident that the interior of the paint layer was relatively compact without any voids present.

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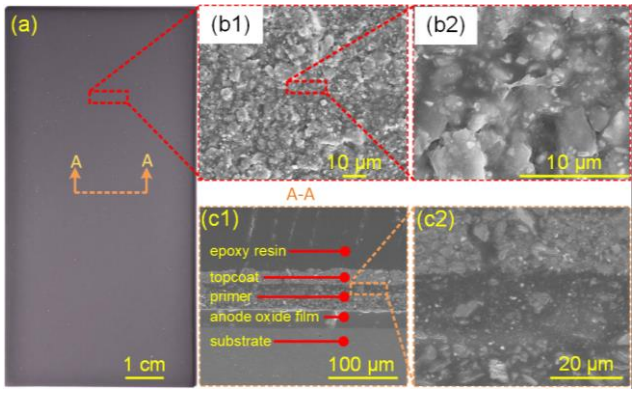

Figure 2 The sample used in the paint removal experiment. (a) Macroscopic photo of sample; (b) SEM images of the paint surface; (c) SEM images of the sample cross-section.

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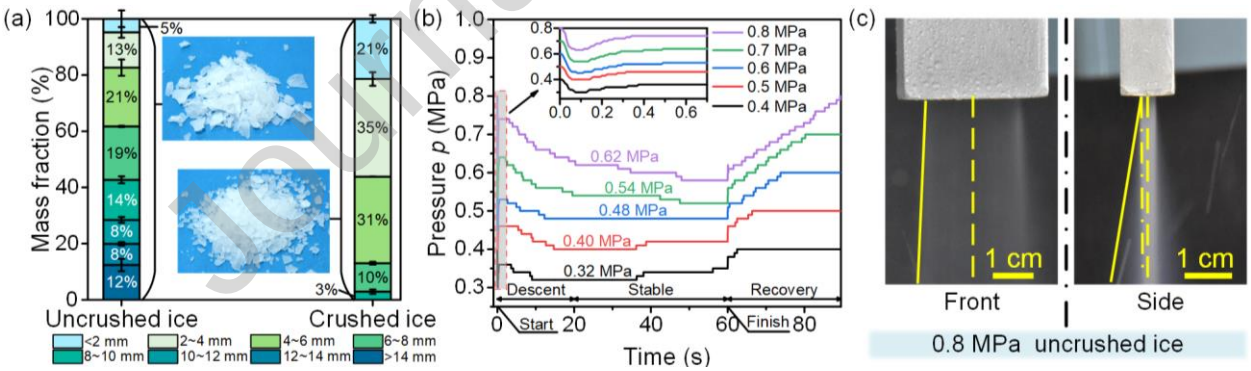

Figure 3 Characteristics of the paint removal system based on the AJUIP: (a) Size distribution of different types of ice particles, (b) Variation of the air pressures in the paint removal system, (c) Photographs of the ice jet in different perspectives.

In order to investigate the impact of the ice particles in different states on the paint removal, the study not only varied the temperatures of the ice particles but also controlled their sizes using an ice crusher. The original ice particles produced by the ice maker were unevenly sized flakes, with a wide range of the particle sizes as depicted in Figure 3(a). The distribution of particle sizes ranged from 2 to 14 mm, and the most abundant sizes fell within the range of 4~8 mm. Following crushing, there were no remaining ice particles larger than 10 mm, and more than half of the remaining ice particles were distributed within the ranges of 2~6 mm.

When the cleaning process started, the high-pressure air stored in the gas container was rapidly released, resulting in a brief pressure drop of approximately 0.1 s, followed by a rapid increase and subsequent gradual decrease in pressure. After about 20 s, the exhaust volume of the air compressor stabilized to match the consumption of the ice jet cleaner, leading to a stable pressure within the gas container at about 80% of its initial level. All experiments were conducted after reaching the stabilized state. Upon completion of the operation, the air pressure within the gas container gradually increased and returned to its initial level after approximately 30 s. The variation of the air pressures within the gas container is illustrated in Figure 3(b).

The distribution of the ice particles at the rectangular nozzle outlet in different perspectives within the air jet is shown in Figure 3(c). In the front view, the jet exhibited a scattered pattern with a scattering angle of about 2°. On the same horizontal plane, the distribution of the ice particles appeared relatively uniform. However, when observed laterally, the jet displayed distinct divergence characteristics. Horizontally divided into two parts, there was a denser and faster region near the axis, which gradually concentrates as the distance from the jet increases, forming an angle of approximately 2° between its boundary and axis. As the distance from the axis increased, there was a decrease in the ice particles, and a divergent trend emerged with an angle of approximately 10° between its boundary and axis. These divergence characteristics suggested that increasing the distance between the nozzle and workpiece could expand the impact area of the ice particle jets; however, this would also reduce the erosion effect on workpieces by individual ice particles. Another region distant from the axis exhibited a lower concentration of the ice particles, which dispersed as the distance from the jet increased. The angle between the boundary of the AJUIP and the axis measured approximately 10°. Based on the divergent characteristics of the AJUIP, it could be inferred that increasing the distance between the nozzle and workpiece might expand the impact area of the AJUIP. However, conversely, this action would also diminish the erosion effect of single ice particle on the workpiece.

3.2 Effects of the AJUIP on the Substrate

To assess the potential impact of the AJUIP on the substrate, an impact test was conducted on the unpainted samples. The experiment utilized uncrushed −86 °C ice particles at an initial pressure of 0.8 MPa and a target distance of 4 cm, with the nozzle positioned at a 45° angle to the sample surface. Surface roughness measurements were taken after various exposure times, and the results are shown in Figure 4(a). The results indicated that both *Sa* and *Sq* decreased following ice spraying, stabilizing after approximately 30 s. However, *Sz* did not exhibit a significant change throughout the testing period.

The cross-sectional comparison of the anodic oxide layer before and after blasting is shown in Figures 4(b), (c). No discernible defects or damages on a macroscopic scale were observed following the impact of the AJUIP. Analysis of the SEM images revealed a reduction in the thickness of the anodic oxide layer from 23.2±1.0 μm to 22.2±0.2 μm, representing a change within 5%. Consequently, it could be considered that the AJUIP exerted negligible influence on the anodic oxide layer present on the sample surface. The elemental distribution characteristics on the substrate surface are presented in Figure 4(d). EDS mapping analysis confirmed that Al, C, O, S, and Si constituted the predominant components of the anodic oxide layer. The two-dimensional elemental mapping demonstrated spatially homogeneous distributions of all detected elements, without observable aggregation phenomena.

As shown in Figure 4(e), following 50 s blasting period, no discernible pits were observed on the sample surface at a macroscopic level, and the surface retained its metallic luster. SEM images revealed that the original sample surface was adorned with numerous micrometer-scale rough structures and pores. Subsequent to 10 s blasting duration, the quantity of rough structures on the surface markedly decreased, while the size of the pores remained unaltered. After 30 s of blasting, the micrometer-scale rough structures on the sample surface had essentially vanished. Upon reaching a blasting time of 50 s, there were still no impact pits on the sample surface, and its microstructure closely resembled that at 30 s. These results suggested that the AJUIP could diminish the roughness of the sample surface and exert a certain polishing effect without inducing plastic deformation to the surface. Therefore, the AJUIP could be considered as non-damaging to the sample substrate.

3.3 Effects of the Blasting Time on Paint Removal

To investigate the shortest time required for the AJUIP cleaning of a specific area, targeted paint removal experiments were conducted with experimental parameters consistent with the previous section. Figure 5(a) illustrates the progression of paint removal on the sample surface at various blasting times. After 10 s of the AJUIP, partial paint peeling occurred, revealing some substrate surface while leaving a substantial amount of flocculent paint residue within the jetting region. Following 20

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seconds of blasting, there was an expansion in the overall outline of the jetting region and a significant reduction in internal residual paint. Continuing to blast, it could be observed that the residual paint region within the jetting region continued to decrease. At 40 s of blasting, a large residue-free region appeared within the jetting region, and after 50 s it was observed that little change occurred in the size of the jetting region, indicating a completion of cleaning at this time.

Chemical element composition analysis was performed on the jetting region, and the EDS test results are shown in Figure 5(c). The presence of elements Na, P, Zn, and Ba in the paint layer was no longer detected after cleaning, indicating the absence of residual paint in the cleaned area. Al and O were identified as the predominant chemical elements in the jetting region, suggesting that the hard anodic oxide layer on the substrate was the primary component, and no other pollutants had been introduced.

To better characterize the cleaning efficacy of the AJUIP, the geometric shape of the cleaning area was defined, as shown in Figure 5(b). The shape of the region cleaned by the AJUIP was an irregular polygon, and the smallest polygon encompassing all exposed substrates was designated as the jetting region (JR). The actual area of the exposed substrate was defined as the bare region (BR). Partially residual paint might be present within the JR, and extending the blasting time could reduce the area of the residual paint. The maximum rectangular area devoid of any paint layer was defined as the residue-free region (RFR). Given that the nozzle outlet shape was rectangular, it was ideal for the shape of the ice particle jet to also be rectangular. Therefore, a minimum rectangle enclosing the entire JR was defined as an envelope rectangle (ER). Additionally, the percentage of the bare region (P_{BR}) was defined to quantitatively assess the cleanliness of the JR:

$$P_{BR} = \frac{A_{BR}}{A_{JR}} \times 100\%, \qquad (1)$$

where A_{BR} and A_{JR} represent the area of the BR and JR.

Figure 5(d) illustrates the variations of the lengths and widths of the ERs with the blasting time. Both dimensions exhibited rapid growth within the initial 30 s of the AJUIP, followed by a deceleration in their expansion rates until reaching maximum area. Throughout the entire jetting process, the variations of the lengths and widths consistently maintained similar magnitudes. The areas of the JR, BR and RFR over time are shown in Figure 5(e). As the blasting time increased, both the areas of the JR and BR expanded. In the early stages of the AJUIP, although some parts of the substrate became exposed, there was still residual paint adhering to the surface. Approximately 0.30 cm^2 RFR emerged after 40 s of the AJUIP, and expanded to 0.65 cm^2 after 50 s. Furthermore, the P_{BR} continued to rise steadily during the jetting process. At 50 s of the blasting, this ratio reached up to 91.6%, indicating that the majority of the paint layers in the JR had been cleaned off by the AJUIP.

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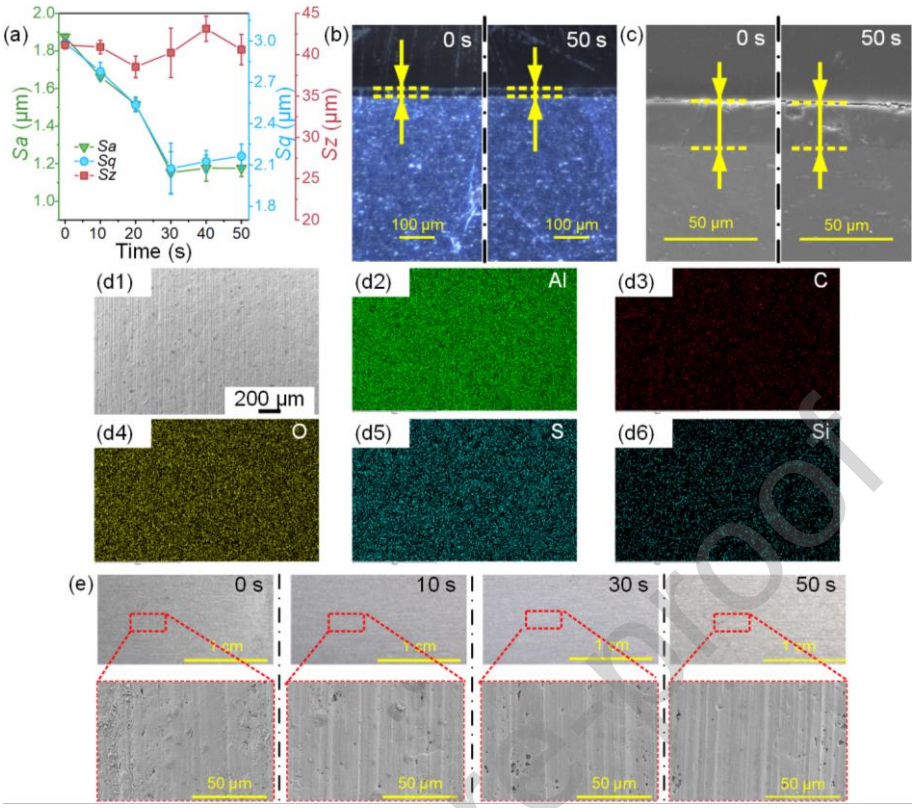

Figure 4 Effects of the AJUIP on the anodic oxide layer: (a) Variation of surface roughness of samples after different blasting times, (b) Cross-sectional photos of samples before and after the AJUIP, (c) SEM images of sample cross-sections before and after the AJUIP, (d) EDS mappings of the sample surface, (e) Photos and SEM images of samples after different blasting times.

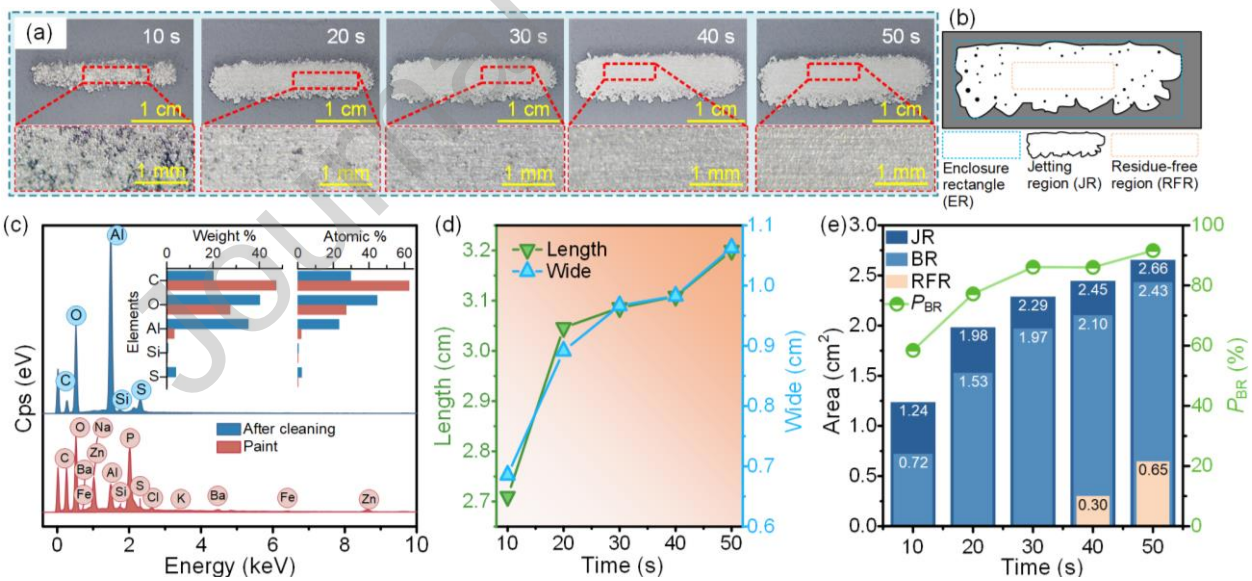

Figure 5 Effects of the blasting time on paint removal: (a) Macroscopic photos of the sample surface at different blasting times, (b) Graphical interpretation of different defined regions, (c) Variations of chemical elements on the sample surface before and after the AJUIP, (d) Variations of the sizes of the ERs at different blasting times, (e) Variations of the area of each region at different blasting times.

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3.4 Effects of the Blasting Pressure on Paint Removal

The high-pressure air served as the carrier of the ice particles, with the pressure directly determining the kinetic energy of the ice particles. Figure 6(a) shows the samples following 50 seconds of blasting at the varying blasting pressures, providing a more intuitive observation of the paint layer peeling process. At the blasting pressure of 0.4 MPa, the kinetic energy of individual ice particle was extremely low, resulting in only tiny peeling of the paint layer after 50 seconds of blasting. Nevertheless, there was virtually no residual paint within the exposed region, presenting as a circular shape with a diameter of about 1 mm. Upon increasing the blasting pressure to 0.5 MPa, there was a slight expansion in the BR on the sample, and it could be observed as a pattern consisting of three stacked circular shapes, each with a diameter of around 1 mm. The cleanliness within each circle was relatively high; however, most residual paint was still present at their boundaries. At a blasting pressure of 0.6 MPa, three larger independent JRs emerged on the sample surface. Within each JR, a substantial amount of flocculent residual paint was present. Although variations in the shapes and sizes among the JRs, it was evident that their edges consisted of multiple arc shapes. Upon further increasing the blasting pressure to 0.7 MPa, only one JR was presented, while the flocculent residual paint persisted. Subsequently, at the blasting pressure of 0.8 MPa, the entire JR exhibited a visualized rectangular shape with minimal residual paint remaining inside.

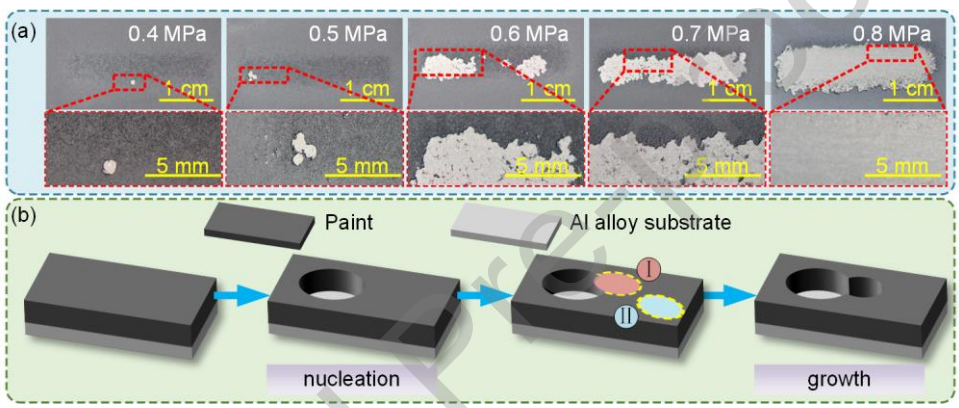

Figure 6 Effects of the blasting pressure on the paint removal: (a) Macroscopic photos of the sample surface under different blasting pressures, (b) Schematic diagram of the paint removal process by the AJUIP.

Based on the aforementioned experimental results, the paint removal process could be divided into four stages. Initially, when an ice particle with high kinetic energy impacted a region where adhesion between the paint layer and substrate was relatively weak, peeling occurred in a circular shape. This initial peeling area was referred to as the "nucleus", and this stage was named as the "nucleation stage". It was worth noting that the "nucleus" seemed to be more inclined to appear in the location near the lower edge within the cleaning area.

After the formation of the "nucleus," the contact length between the paint layer in its vicinity (region I in Figure 6(b)) and the rest of the paint layer was shorter than that between the paint layer located far away from the "nucleus" (region II in Figure 6(b)) and the rest of the paint layer. As a result, there was a reduced bonding force between the paint layer in region I and the overall paint layer, rendering these regions more susceptible to peeling off from the substrate upon impact from the ice particles.

When a greater number of ice particles impacted the paint layer, the regions surrounding the "nucleus" would peel off successively, manifesting the growth of the "nucleus", thus defining this stage as the "growth stage". It was critical to note that the expansion of the "nucleus" was not a smooth continuous process due to peeling occurring in units of one circular paint layer at a time. The impact points of the ice particles were at the center of each circular paint layer, resulting in minimal force exerted on the outer edges and leading to partial paint remaining in these regions, namely, flocculent paint residue. The "nucleus" continued to expand and evolved into a larger "island" region. Both "nucleation stage" and "growth stage" might occur in multiple locations within the JR, giving rise to several "island" regions. As each individual region expanded, inevitable fusion occurred between them, forming a larger “continent”. This stage was called “fusion stage”. At this point, only the largest “continent” remained within the JR while numerous flocculent residues persisted within it. The continued impact of the ice particles on the workpiece surface effectively removed most remaining paint and ultimately eliminated all paint within that region – referred to as “cleaning stage”.

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3.5 Effects of the Blasting Distance on Paint Removal

Figure 7(a) shows the paint removal effect at different distances from the nozzle to the workpiece surface. As the blasting distance increased, there was a trend of initially increasing and then decreasing in the geometric size of the JR. When the distance was 4 cm, the size of the ER reached its maximum value. This was because the ice particles would diverge after leaving the nozzle. A closer jet distance resulted in a more concentrated ice particle jet, leading to higher cleanliness in the JR, while the area of this region was limited. Properly increasing the distance could expand the JR of the AJUIP while maintaining high cleanliness levels. However, excessively increasing the distance might cause dispersion of the ice particles and a decrease in ice particle quantity per unit area. Furthermore, when the ice particles were too far from the nozzle, it reduced the acceleration effect of the compressed air on them, resulting in reduced impact force on paint layers. At a distance of 10 cm, the sample surface even retained two "islands". Therefore, the larger the distance, both the size of ER as well as the areas of JR and BR all decreased, as shown in Figure 7(b), (c).

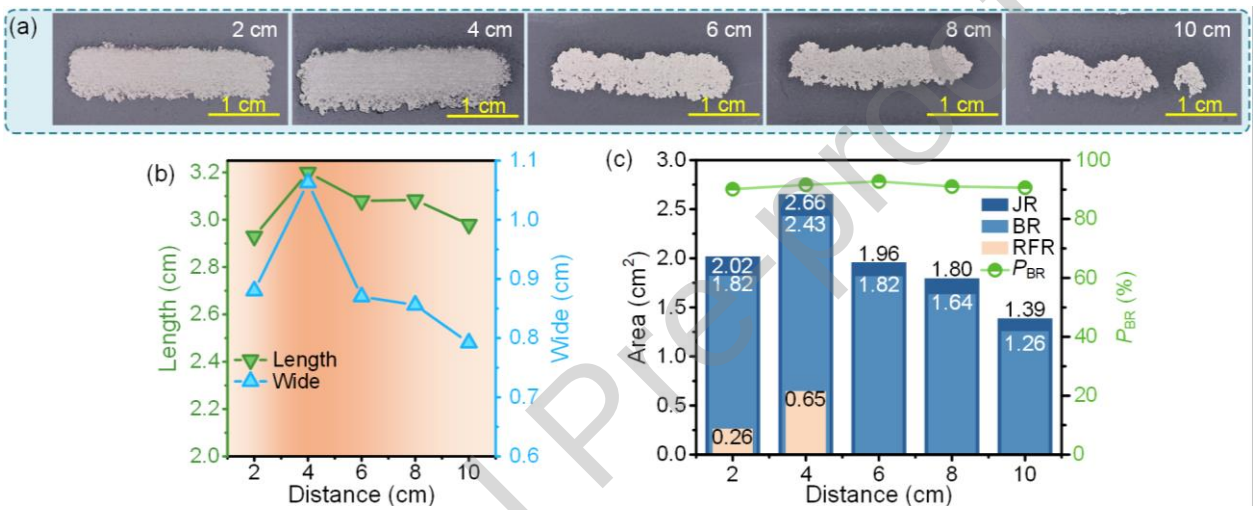

Figure 7 Effects of the blasting distance on the paint removal. (a) Macroscopic photos of the sample surface under different distances; (b) Variations of the sizes of the ERs at different blasting distances; (c) Variations of the area of each region at different blasting distances.

3.6 Effects of the Blasting Angle on Paint Removal

Figure 8(a) shows the paint removal effect at various angles between the nozzle and the workpiece surface. It was evident that when the blasting angle was greater than 45°, a larger area of JR with litter residual paint was obtained, thus achieving a high level of cleanliness. Conversely, when the blasting angle decreased to 30°, there was a notable reduction in JR and a large amount of residual paint inside this region. Furthermore, it was observed that at a perpendicular angle (i.e., 90°), both the outlines of the upper and lower edges of the JR exhibited similar roughness. As the angle decreased, the outline of the upper edge became smoother while the roughness of the lower edge remained unchanged. The reason for this phenomenon was that the impact of the ice particles on the paint layer was not entirely composed of the impact force perpendicular to the surface of the workpiece; there was also a downward shear force along the surface of the workpiece. This phenomenon resulted in differences in contours on two edges as an arc-shaped tearing occurred on the lower edge paint layer under shear force and high-speed airflow, while less influence was exerted on the upper edge paint layer by shear force. When the blasting angle decreased to 30°, ineffective removal of surface paint layer occurred as the force component from the ice particles impact became lower than the shear force component.

From Figure 8(b), it was evident that the length of the ER remained relatively constant within the angle ranging from 90° to 45°, while its width increased exponentially as the angle decreased. This phenomenon could be attributed to the tilt of the sample along the length direction of the envelope rectangle, resulting in an extension of the width, as shown in Figure 8(c). Assuming that at an angle of 90° between the nozzle and workpiece surface, the width of the ER was l_0, then for any α (α>45°), we could express the width l of the ER as follows.

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$$l = \frac{l_0}{\cos(\frac{\pi}{2} - \alpha)}. \tag{2}$$

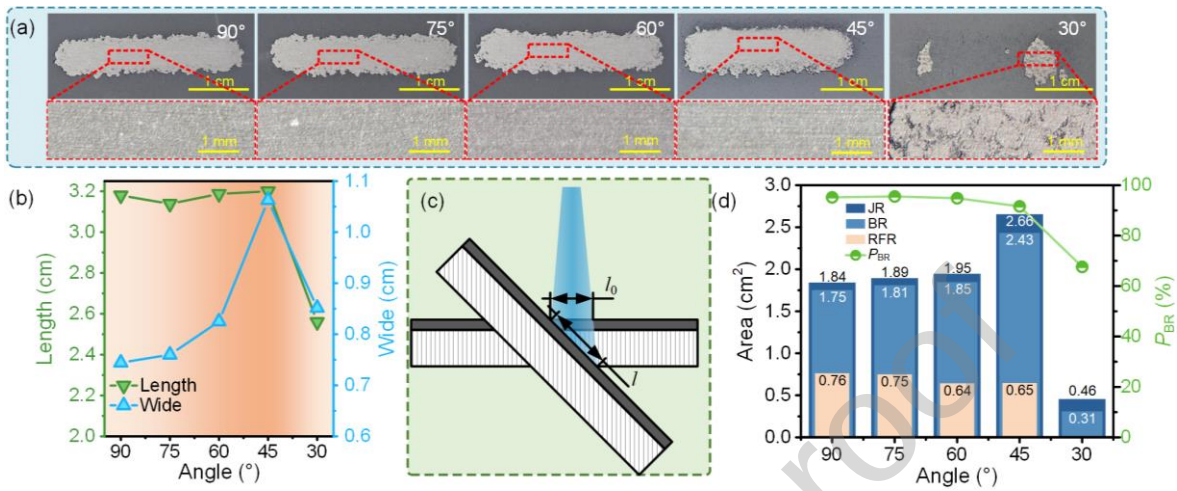

Figure 8 Effects of the blasting angle on the paint removal: (a) Macroscopic photos of the sample surface under different blasting angles, (b) Variations of the sizes of the ERs at different blasting angles, (c) Diagram of workpiece rotation, (d) Variations of the area of each region at different blasting angles.

Given the apparent divergence in the movement trajectory of the ice particles upon leaving the nozzle, it followed that the dimensions of the ER at positions further from the nozzle would also be proportionally larger. Therefore, it became necessary to introduce a divergence factor k into Eq. (2):

$$l = k\frac{l_0}{\cos(\frac{\pi}{2} - \alpha)}, \tag{3}$$

where k>1 and is related to the distance.

Similarly, the area of the JR and BR also increased proportionally, as shown in Figure 8(d). Consequently, tilting the workpiece at a certain angle could effectively expand the cleaning area of the AJUIP and thereby improve cleaning efficiency. Within the range of 90° to 45° angles, the P_{BR} consistently exceeded 90%.

3.7 Effects of the Ice Particle Size on Paint Removal

Figure 9(a) compares the cleaning effects of ice particles of different sizes on paint. The results indicate that ice particles of two distinct sizes produced similar cleaning pattern outlines. Based on the dimensions of BR provided in Figure 9(b), the crushed ice particles lead to reduced length and width of the ER, suggesting that smaller ice particles achieve higher jet concentration. This could be attributed to the higher inertia of larger ice particles, which made them less susceptible to alterations in their trajectories by airflow. In contrast, the smaller particles were more easily manipulated by the air jet, resulting in a narrower and more concentrated jet beam with reduced dispersion. However, due to their limited size, the crushed particles struggled to effectively impact all areas within the JR, leading to considerable flocculent residual paint debris. Consequently, both in terms of cleaned area or overall cleanliness, the performance of smaller ice particles was inferior to that of larger ice particles, as shown in Figure 9(c).

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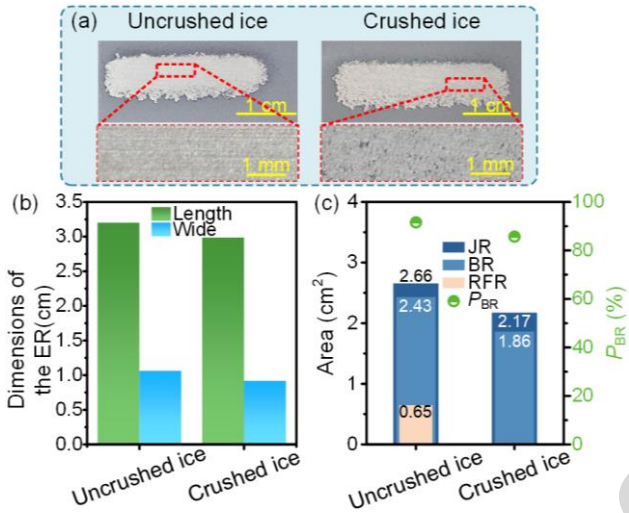

Figure 9 Effects of the ice particle sizes on the paint removal: (a) Macroscopic photos of the sample surface under different ice particle sizes, (b) Variations of the sizes of the ERs at different ice particle sizes, (c) Variations of the area of each region at different ice particle sizes.

3.8 Effects of the Ice Particle Temperature on Paint Removal

The cleaning effectiveness of the ice particle jet was directly influenced by the hardness of the ice particles, which could be tuned by their temperature. Figure 10(a) shows the varying cleaning effects on paint when using the ice particles at different temperatures. It was observed that utilizing −2 °C ice particles with a Mohs hardness of 2 to clean the sample surface, there were still large areas of uncleansed regions within the JR. Lowering the temperature of the ice particles to −28 °C with a Mohs hardness of 2.5 resulted in an improved cleanliness within the JR, although there were still scattered small amounts of residual paint. Experimental findings indicated that utilizing the ultrahard ice particles not only expanded the JR of the AJUIP, as depicted in Figure 10(b), (c), but also enhanced the cleanliness within this region. Consequently, substituting ordinary ice particles with the ultrahard ice particles significantly enhanced the performance of the ice particle jet technology.

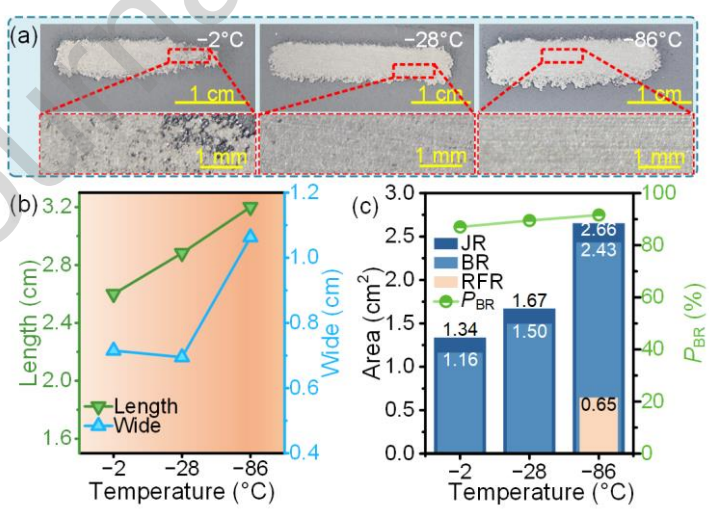

Figure 10 Effects of the ice particle temperatures on the paint removal: (a) Macroscopic photos of the sample surface under different ice particle temperatures, (b) Variations of the sizes of the ERs at different ice particle temperatures, (c) Variations of the area of each region at different ice particle temperatures.

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3.9 Morphological Analysis of the Typical JR

The morphological characteristics of the typical JR were analyzed under the conditions of 0.8 MPa blasting pressure, 45° blasting angle, 4 cm blasting distance, and −86 ℃ ice particle temperature. The surface morphologies of the upper edge, lower edge, and RFR were shown in Figure 11(a). It was evident that a transition region existed between the paint layer and the substrate surface near the upper edge of the JR. The paint layer in this region exhibited peeling in multiple layers. On the surface of the primer, indentations resulting from impact by the ice particles were visible, as shown in Figure 11(b), indicating that the primer had undergone plastic deformation due to the impact. In contrast, a distinct boundary between the paint layer and substrate was observed near the lower edge of the JR, suggesting simultaneous detachment of both primer and topcoat from the substrate surface upon impact by the ice particles. The straight boundaries observed implied that they were not solely formed by the ice particles impact force, but also indicated tearing effects on paint layers from high speed air, as well as shear force from the ice particles during the paint removal process.

The cross-sectional profiles of the upper and lower edges of the JR were depicted in Figure 11(c). It was evident from the figure that the variation in paint layer thickness at the upper edge exhibited a distinct "step-like" pattern, characterized by pits resulting from impact of the ice particles on the primer surface. Conversely, at the lower edge, there was an abrupt drop to zero thickness, presenting a "cliff-like" shape.

The elemental distribution results of the surface in the cleaning area are shown in Figure 11(d). The sample surface after ice jet treatment is highly consistent with the original untreated one (Figure 4(d)) in elemental distribution. EDS mappings revealed that the elemental types of the two samples' surfaces were similar, and the two-dimensional spatial distribution of each element remained uniform. This indicated that AJUIP process caused no obvious elemental diffusion or introduction of contaminants.

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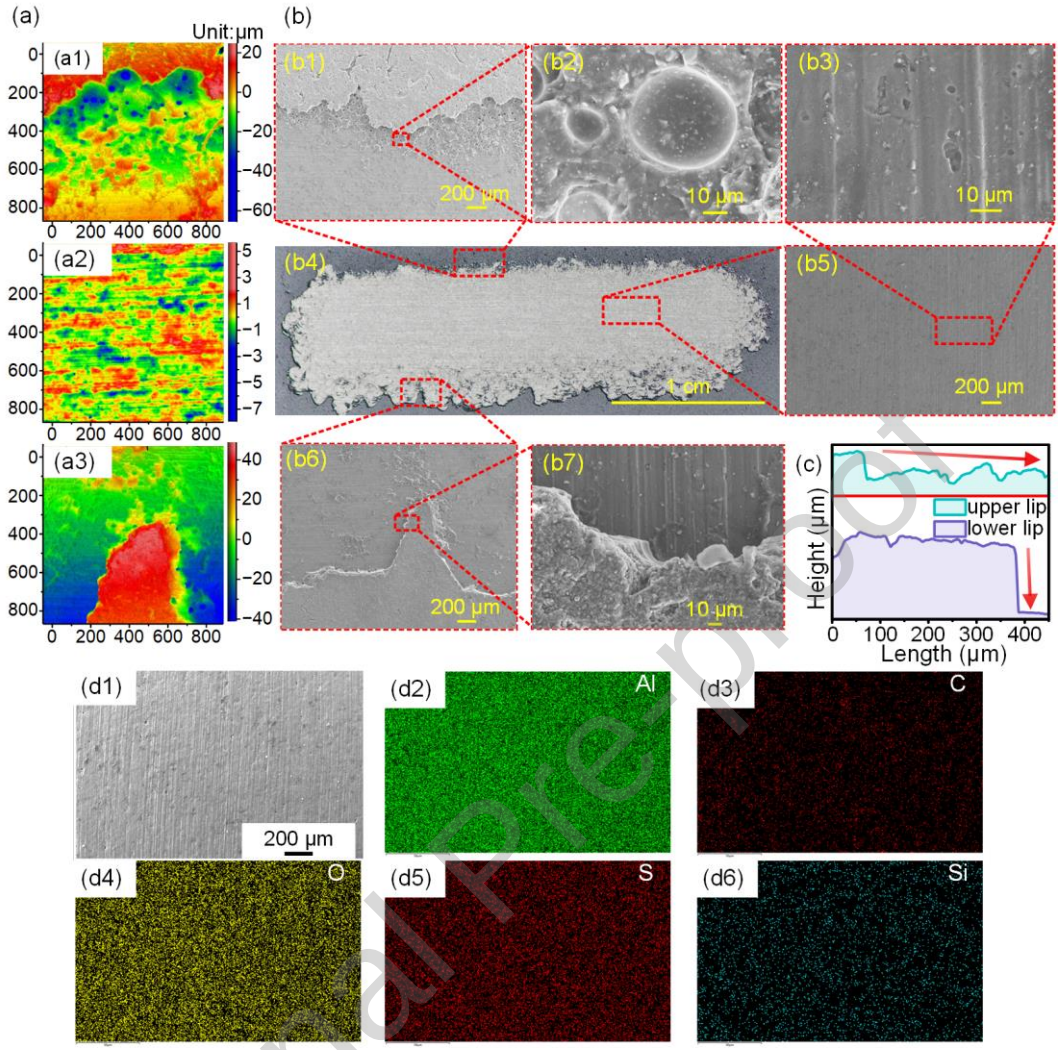

Figure 11 Morphological analysis of the typical JR: (a) Microscopic morphologies of different regions, (b) SEM images of different regions, (c) Cross-sectional profiles of different regions, (d) EDS mappings of the cleaned region.

3.10 Analysis of Paint Removal Mechanism of the AJUIP

In order to explain the discrepancy in morphologies between the upper and lower edges of the JR, the FLUENT software was utilized to simulate the flow field within the region spanning from the nozzle to the workpiece. The pressure at the inlet of the nozzle was set at 0.62 MPa, which is the pressure after the gas field stabilized under an initial pressure of 0.8 MPa, with a distance of 4 cm between the nozzle and workpiece. The gas field distribution was computed for various blasting angles. Figure 12(a) shows the velocity isograms of gas under the conditions of workpiece at various blasting angles. The gas flow velocity along the axis of the nozzle remained relatively constant at different angles, while a notable disparity in gas velocity was observed on the surface of the workpiece. When the nozzle was perpendicular to the workpiece surface, a symmetrical flow field structure was formed, with high-speed gas flowing along the workpiece surface to both sides after impacting. The gas velocity was at its minimum at the position directly facing the nozzle, and it progressively rose as the distance from the position increased. Upon reaching its maximum, the gas velocity subsequently diminished. However, when the workpiece was tilted at a certain angle, an asymmetrical distribution of the gas field on the workpiece surface emerged. Gas moving downwards along the workpiece surface from the center could reach higher velocities, while gas moving in the opposite direction had lower velocities. Furthermore, as the blasting angle of the workpiece increased, there was a gradual increase in the difference between maximum velocities on both sides, as shown in Figure 12(b). The presence of this difference was responsible for the distinct morphologies of the edges flanking the JR. Furthermore, the phenomenon of

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higher gas velocity in the location on the bottom half of the JR could also account for why "nucleus" consistently emerged first in this particular region during the nucleation stage.

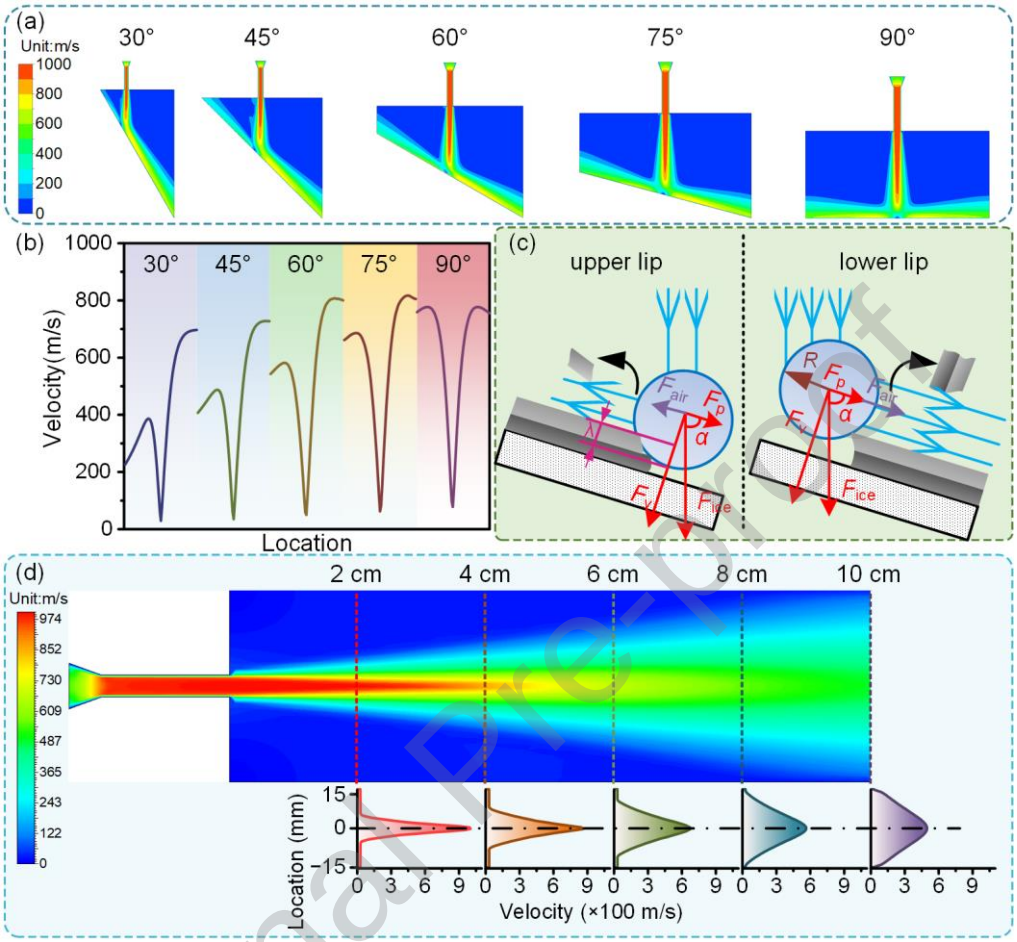

Figure 12 Analysis of paint removal mechanism of the AJUIP: (a) Gas velocity isograms at different angles, (b) Gas velocity on the surface of the workpiece at different angles, (c) Schematic diagram of paint removal by the ice particle jet, (d) Velocity isograms of internal and external flow fields in the nozzle and flow velocities at different distances.

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The damage caused by the AJUIP to the paint layer was categorized into two main components: the impact of the ice particles and the blowing effect of high-speed gas. The impact of the ice particle on the paint layer was further classified into vertical impact force (F_v) and parallel shear force (F_p) to the workpiece surface. Figure 12(c) illustrates a schematic diagram depicting an ice particle impacting the paint layer. The radius, density, velocity, and impact depth of the ice particle were denoted as R, ρ, v, and λ, respectively. According to Hertz's elastic contact theory [24], λ could be determined from Eq. (4).

$$\lambda = \left[\frac{9}{16} \frac{F_v^2}{R} \left(\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2} \right)^2 \right]^{\frac{1}{3}}, \qquad (4)$$

where E_1 and ν_1 are the elastic modulus and Poisson's ratio of the paint layer, and E_2 and ν_2 are the elastic modulus and Poisson's ratio of the ice particles.

Rewriting Eq. (4) yields Eq. (5):

$$F_v = \frac{4}{3} \left(\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2} \right)^{-1} R^{\frac{1}{2}} \lambda^{\frac{3}{2}}. \qquad (5)$$

According to Newton's second law, $mv\mathrm{d}v = -F_v v\mathrm{d}t$ is obtained. By substituting Eq. (5) into this equation and integrating over the process of the ice particles impacting the paint layer, we obtain:

$$\int_{v_0}^{0} mv\mathrm{d}v = \int_{0}^{t_0} -F_v v\mathrm{d}t = -\int_{0}^{\lambda_0} \frac{4}{3}\left(\frac{1-v_1^2}{E_1}+\frac{1-v_2^2}{E_2}\right)^{-1} R^{\frac{1}{2}}\lambda^{\frac{3}{2}}\mathrm{d}\lambda, \tag{6}$$

where v_0 is the velocity component of the ice particles perpendicular to the workpiece surface, and t_0 is the duration of the impact process. After rearranging, we get:

$$\lambda = \left[\frac{5\pi}{4}\rho\left(\frac{1-v_1^2}{E_1}+\frac{1-v_2^2}{E_2}\right)\right]^{\frac{2}{5}} R v_0^{4/5}. \tag{7}$$

According to Eq. (7), we inferred that the impact depth λ of a single ice particle impacting on the workpiece might be proportional to the radius R and $v_0^{4/5}$ of the ice particle. When λ exceeded the thickness of the paint layer, the paint was susceptible to cracking and detachment from the workpiece due to force exerted by the AJUIP.

On the other hand, the paint layer undergoes shear wear induced by the shear force F_P generated by ice particles. Finnie investigated the plastic shear of abrasive materials and proposed a material removal model for plastic shear wear [25]. Assuming the paint layer is smooth and the ice particles are rigid particles, when an ice particle impacts the coating at velocity v with an incident angle α (neglecting particle rotation), the volume removed by shear wear can be expressed as:

$$V_P \approx \begin{cases} \dfrac{mv^2}{8P}(\sin 2\alpha - 3\sin^2\alpha), & \alpha \le 18.5°, \\ \dfrac{mv^2}{24P}\cos^2\alpha, & \alpha \ge 18.5°, \end{cases} \tag{8}$$

where V_P is the volume removed by shear wear, and P is the flow stress of the paint. Due to the inherent surface roughness of practical paint layer, the actual material removal volume typically exceeds theoretical predictions.

According to established abrasive wear theory, v of ice particles predominantly influenced the paint removal efficiency of the AJUIP. The pressure of the compressed air directly governed the initial velocity of ice particles exiting the nozzle. However, as the blasting distance increased, the particle velocity progressively decayed, thereby diminishing the coating removal efficacy.

Regardless of the position of impact of the ice particle on the paint, the direction of the parallel component F_t of the impact force F_{ice} was always downward towards the surface of the workpiece. However, the direction of the force F_{air} from high-speed flowing air on the paint layer varied with its position. In the upper part of the JR, F_{air}'s direction was upward along the workpiece surface; whereas in the lower part of the JR, its direction was downward along the workpiece surface. Hence, the resultant force F_1 exerted on the paint layer at the upper boundary of the JR was $F_1 = F - F_P$, whereas the resultant force F_2 acting on the paint layer at the lower boundary was $F_2 = F + F_P$. The value of F_1 was comparatively smaller, thereby selectively removing only the topcoat with lower adhesion strength and preserving the primer with higher adhesion in that specific region; conversely, F_2 had a larger value, leading to complete removal of both primer and topcoat in the lower region of the JR.

Through the aforementioned analysis, it was evident that tilting the workpiece at a specific angle could expand the cleaning area during paint removal. However, excessive tilt might have resulted in a reduced vertical velocity component F_v of the ice particle, leading to inadequate impact depth λ of it for reaching the contact position between the paint layer and the workpiece. Consequently, this would have hindered effective paint layer removal. Furthermore, by adjusting both velocity and size of the ice particle for precise control of λ, it was anticipated to achieve selective removal of multi-layer coatings.

The velocity distribution isogram of the gas field inside and outside the nozzle is shown in Figure 12(d). It was evident from the figure that the overall distribution of the gas field exhibited divergence. Within a jet distance range of 0~5 cm, there was a concentration of high velocity gas near the axis. When combined with Figure 3 (c), it could be inferred that this region also corresponded to where the ice particles were concentrated, indicating that this region represented the highest cleaning efficiency for the AJUIP. As the blasting distance increased, the gas field gradually dispersed and peak velocity decreased, diminishing its acceleration effect on the ice particles and causing them to disperse as well; simultaneously, reduced gas velocity weakened its sweeping effect on paint layers. The combination of these factors resulted in significantly less effective paint removal by the AJUIP at longer distances compared to an optimal range.

4 Conclusions

We proposed a novel paint removal technique utilizing the air jet with ultrahard ice particles (AJUIP), and conducted

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removal experiments on the epoxy organic silicon heat-resistant paint from aluminum (Al) alloy surface. The results of the paint removal experiments and simulation demonstrated that:

(1) Following the application of the AJUIP onto the substrate surface for 50 s, there was a certain reduction in surface roughness of the sample, with no apparent macroscopic or microscopic defects and damage observed, and virtually no change in thickness and morphology of the anodic oxide layer. Therefore, it could be considered that the AJUIP was non-destructive to the sample surface.

(2) The blasting time, pressure, distance from the surface, and blasting angle of sample were all influential factors on the efficacy of paint removal. Prolonging the blasting time and increasing the pressure led to a wider cleaning region and improved cleanliness. The optimal blasting distance and blasting angle of sample enhanced the effectiveness of the AJUIP in removing paint. The temperature and size of the ice particles played a crucial role in determining the effectiveness of paint removal. Lower temperature resulted in higher hardness of the ice particles, while larger-sized ice particles could create deeper cracks, both contributing to superior paint removal effects.

(3) The removal process of paint layer by the AJUIP could be divided into four stages: nucleation stage, growth stage, fusion stage, and cleaning stage. The peeling of the paint layer was carried out in units of circular with a diameter of about 1 mm. The force exerted by the AJUIP on the paint layer was categorized into impact force and shear force. Effective removal of the paint layer only occurred when the crack length caused by the impact force exceeded the thickness of the paint layer. In conclusion, the AJUIP was proved to be a more efficient method for paint removal compared to a normal ice particle jet, with potential for achieving selective coating removal through manipulation of physical properties of the ice particles.

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Author contributions

Jinlong Song was in charge of the framework design of manuscript; **Bingzhen Zhang** was in charge of the experiments, and wrote the original manuscript; **Xiansheng Zhen** was in charge of the samples required for the experiments; **Xiaoqing Cui** assisted with experiments; **Yang Chen** assisted in revising manuscripts; **Anqin Zhang**, **Liling Dan**, **Youtao Wang**, **Lei Jin** assisted with sampling. All authors read and approved the final manuscript.

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Data availability

The data will be made available on request.

Declarations

Competing Interests

The authors declare no competing financial interests.

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References

[1] Z G Zhu, Z H Hu, H L Seet, et al. Recent progress on the additive manufacturing of aluminum alloys and aluminum matrix composites: Microstructure, properties, and applications. *International Journal of Machine Tools and Manufacture*, 2023, 190: 104047.

[2] D F Yan, J Y Lin, Y Chen, et al. High-efficiency water collection of superhydrophobic condensation absorber. *Advanced Science*, 2025, 12(13): 2417024.

[3] T Trzepiecinski, S M Najm. Current trends in metallic materials for body panels and structural members used in the automotive industry. *Materials*, 2024, 17 (3): 590.

[4] S S Li, X Yue, Q Y Li, et al. Development and applications of aluminum alloys for aerospace industry. *Journal of Materials Research and Technology*, 2023, 27: 944-983.

[5] D F Yan, Y Lu, J Y Lin, et al. Enhancing water transportation capacity by asymmetrical patterned surface with super-wettability. *Applied Physics Letters*, 2024, 125(7): 071601.
[6] X F Yu, G Q Zhang, Z X Zhang, et al. Research on corrosion resistance of anodized and sealed 6061 aluminum alloy in 3.5 % sodium chloride solution. *International Journal of Electrochemical Science*, 2023, 18(5): 100092.
[7] Y Y Ji, Q Hu, D H Xia, et al. Corrosion susceptibility of passive films on 1060, 2024, and 5083 aluminum alloys: Experimental study and first-principles calculations. *Journal of the Electrochemical Society*, 2023, 170(4): 041505.
[8] Y Chen, S Y Li, Z Y Liu, et al. Anticorrosion property of alcohol amine modified phosphoric and tannic acid based rust converter and its waterborne polymer-based paint for carbon steel. *Coatings*, 2021, 11(9): 1091.
[9] J Machotova, H Zgoni, S Podzimek, et al. Property study of structured self-crosslinking acrylic latex binder: Effect of molar mass and particle design. *Progress in Organic Coatings*, 2017, 111: 258-266.
[10] M C Mejía, J Palacio, E A Murillo. Comb-shaped silicone-alkyd resins with high solid content. *Progress in Organic Coatings*, 2017, 105: 336-341.
[11] S Sogukkanli, M Yilmazoglu, M A Tasdelen, et al. Hybrid film properties of the linseed oil based alkyd resin modified with glycidyl polyhedral oligomeric silsesquioxane. *Progress in Organic Coatings*, 2018, 124: 175-184.
[12] H H Wang, R Y Guo, Y D Shen, et al. Waterborne polyaniline-graft-alkyd for anticorrosion coating and comparison study with physical blend. *Progress in Organic Coatings*, 2019, 126: 187-195.
[13] G Sheoran, S Sharma, C Shakher. Monitoring of drying process and cracking/disbonding of paints using lensless Fourier transform digital holography. *Optics and Lasers in Engineering*, 2011, 49(1): 159-166.
[14] X F Yang, D E Tallman, G P Bierwagen, et al. Blistering and degradation of polyurethane coatings under different accelerated weathering tests. *Polymer Degradation and Stability*, 77(1): 103-109.
[15] S Xiong, X J Jia, S S Wu, et al. Parameter optimization and effect analysis of low-pressure abrasive water jet (LPAWJ) for paint removal of remanufacturing cleaning. *Sustainability*, 2021, 13(5): 2900.
[16] N Careddu, D Cadeddu, F Sitzia. Graffiti removal from granite cladding by abrasive water-jet. *Journal of Building Engineering*, 2023, 78: 107626.
[17] Z Kuang, W Guo, J N Li, et al. Nanosecond fibre laser paint stripping with suppression of flames and sparks. *Journal of Materials Processing Technology*, 2019, 266: 474-483.
[18] Z J Wang, Z W Chen, H Kang, et al. Investigation of paint removal by atmospheric pressure plasma jet. *Plasma Science and Technology*, 2023, 25(8): 085506.
[19] C N Young, C R Clayton, J P Yesinowski, et al. Physicochemical investigation of chemical paint removers: Interactions of methylene chloride with polyurethane coatings. *Progress in Organic Coatings*, 2014, 77(1): 232-241.
[20] L Yang, Y F Hu, Y Shen, et al. Comparative study of paint removal processes for aluminum alloy skin surface. *Plating and Finishing*. 2024, 46 (5), 1-10. (in Chinese)
[21] D Y Li, X Duan, X Dong. The experimental study on paint removal from aluminum surface using ice jet. *International Conference on Advanced Design and Manufacturing Engineering*, Anshan, Peoples R China, July, 13-14, 2013: 732-736.
[22] A Prezhbilov, M Burnashov, Y Stepanov. Study of the influence of technological, structural and operational parameters on the efficiency of the water-ice jet cleaning process. *IOP Conference Series.: Materials Science and Engineering*, 2020, 709 (4): 44085.
[23] Z P Li, Y Liu, J P Wei, et al. Molecular dynamics simulation and experimental study into effect of temperature on hardness of ice abrasives. *Frontiers in Earth Science*, 2023, 10: 1114421.
[24] Y Zhou, D P Liang. Study on the theoretic model of ultrasonic polishing. *China Mechanical Engineering*. 2005, 16 (8): 664-6. (in Chinese)
[25] C G Zhang, Y Zhang, F H Zhang, et al. Study on removal model of abrasive waterjet machining. *Journal of Mechanical Engineering*. 2015, 51(7): 188-196. (in Chinese)

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